

Multijet balance: smearing the recoil in MC

Four configurations, $R = 0.4$

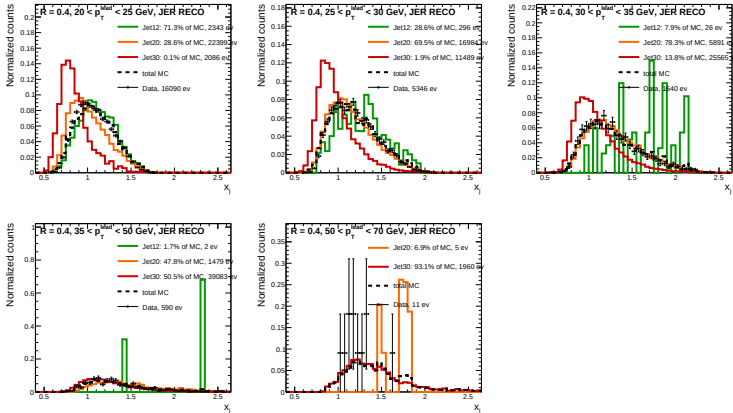
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Common setup

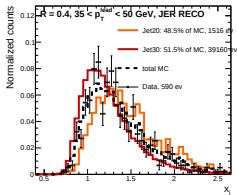
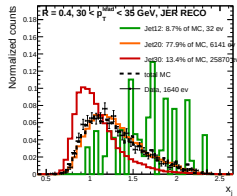
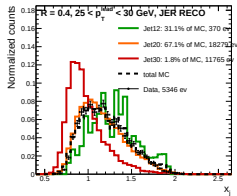
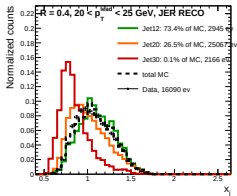
- ▶ $x_j = p_{T,\text{lead}}/|\vec{p}_{T,2} + \vec{p}_{T,3}|$; the recoil is $|\vec{p}_{T,2} + \vec{p}_{T,3}|$.
- ▶ Reprocessed data and Pythia8 trees (new skim, per-event seeded JER smearing).
- ▶ MC: Jet12, Jet20, Jet30; Jet12 starts at truth $p_T = 0$.
- ▶ Leading reco jet capped at 35 GeV in Jet12 and 50 GeV in Jet20.
- ▶ Leading jet ≥ 20 GeV, jets 2 and 3 ≥ 7 GeV each, $|\eta| < 0.7$, $\Delta\phi_{12} > 3\pi/4$, $\Delta\phi_{13} > \pi/2$.
- ▶ Leading- p_T bins 20–25, 25–30, 30–35, 35–50 GeV.
- ▶ MC reweighted in leading p_T and z vertex; two passes.
- ▶ Only the treatment of the MC recoil changes between the configurations.

Why the leading reco jet is capped in Jet12 and Jet20



- ▶ $R = 0.4$ MC by sample, before the caps, with a 50–70 GeV bin.
- ▶ 35–50 GeV: Jet12 is 1.7% of the MC from 2 events ($x_j \approx 1.4, 2.3$).
- ▶ 50–70 GeV: Jet20 is 6.9% of the MC from 5 events ($x_j = 1.5\text{--}1.8$); data has 11 events.
- ▶ The spikes are a few events with large weights.
- ▶ Caps: leading reco jet $\leq 35 \text{ GeV}$ in Jet12, $\leq 50 \text{ GeV}$ in Jet20; the 50–70 GeV bin is dropped.

After the caps (configuration 4)



- ▶ The 35–50 GeV bin is now Jet20 and Jet30 only and is smooth.
- ▶ Jet12 still contributes 8.7% at 30–35 GeV from 32 events.
- ▶ Those events are spiky, but the total MC still follows the data.

The four configurations

	jets 2 and 3	recoil smearing	recoil cut
1	each smeared on its own	none	none
2	each smeared on its own	none	≥ 14 GeV
3	added unsmeared	sum smeared once, every event	≥ 14 GeV
4	one truth jet: added unsmeared otherwise: each smeared on its own	one truth jet: sum smeared once otherwise: none	≥ 14 GeV

- ▶ In configuration 3 the smearing width is taken at the unsmeared recoil p_T , so truth is used only for the sample stitching.
- ▶ Configuration 4 smears the recoil only when jets 2 and 3 belong to a single truth jet (next slide), with the width at that truth jet's p_T .
- ▶ Its other events smear jets 2 and 3 individually, as in configuration 2.
- ▶ The 7 GeV jet cuts use the values before any recoil smearing.

Configuration 4: are jets 2 and 3 one truth jet?

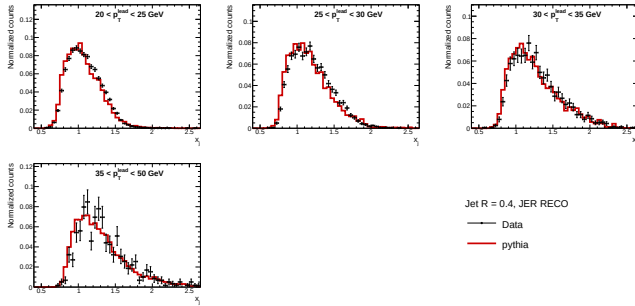
- ▶ Truth jets ≥ 7 GeV; reco jets ranked by calibrated p_T ; matching radius $0.75R$.
- ▶ The leading jet matches a truth jet T_L (stored match, else the nearest truth jet).
- ▶ Jet 2 matches its nearest truth jet $T_R \neq T_L$.
- ▶ Jet 3 matches T_R or no truth jet, never T_L .
- ▶ Every other truth jet is more than $1.5R$ from jets 2 and 3.
- ▶ The summed recoil $\vec{p}_{T,2} + \vec{p}_{T,3}$ lies within $0.75R$ of T_R .
- ▶ If all hold, the sum is smeared once with the width at $p_T(T_R)$, and jets 2 and 3 are scaled to the smeared recoil.
- ▶ Otherwise jets 2 and 3 are smeared individually.
- ▶ At $R = 0.4$ about 32% of the weighted selected MC gets the recoil smearing (22% with truth jets ≥ 5 GeV).

Result: data/MC of $\langle x_j \rangle$, $R = 0.4$

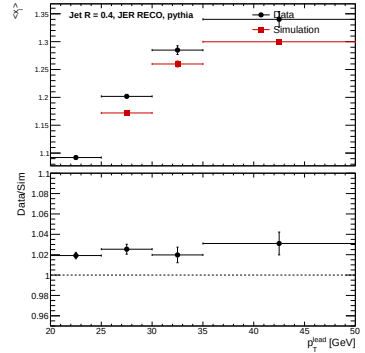
	20–25	25–30	30–35	35–50
1: individual, no recoil cut	1.019 ± 0.003	1.025 ± 0.005	1.020 ± 0.008	1.031 ± 0.011
2: individual, recoil cut	1.026 ± 0.003	1.031 ± 0.005	1.020 ± 0.008	1.030 ± 0.011
3: recoil smeared, all events	0.998 ± 0.004	0.990 ± 0.005	0.979 ± 0.007	0.980 ± 0.011
4: smeared if one truth jet	0.999 ± 0.003	0.995 ± 0.005	0.981 ± 0.008	0.988 ± 0.011
4, truth jets ≥ 5 GeV	1.006 ± 0.003	0.999 ± 0.005	0.990 ± 0.008	0.993 ± 0.011
hybrid: 3 with truth width	1.000 ± 0.004	0.991 ± 0.005	0.979 ± 0.007	0.980 ± 0.011
data events	16090	5346	1640	590

- ▶ The recoil cut removes about 1.5% of data events and moves the ratio by at most 0.007.
- ▶ Smearing the summed recoil once raises the MC $\langle x_j \rangle$ by 2.5–5%.
- ▶ Hybrid (configuration 3 with the truth width in two-truth-jet events) matches configuration 3, so the width choice does not matter.

Configuration 1: individual smearing, no recoil cut

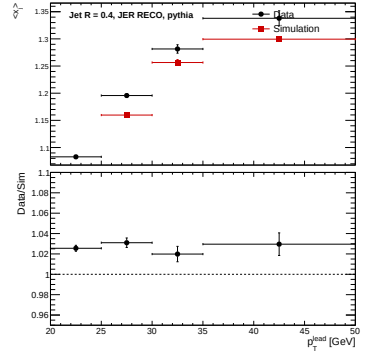
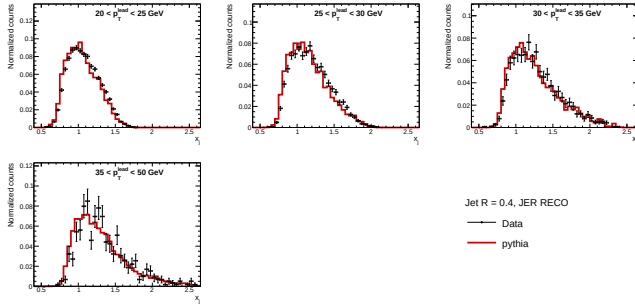


Jet R = 0.4, JER RECO
— Data
— pythia



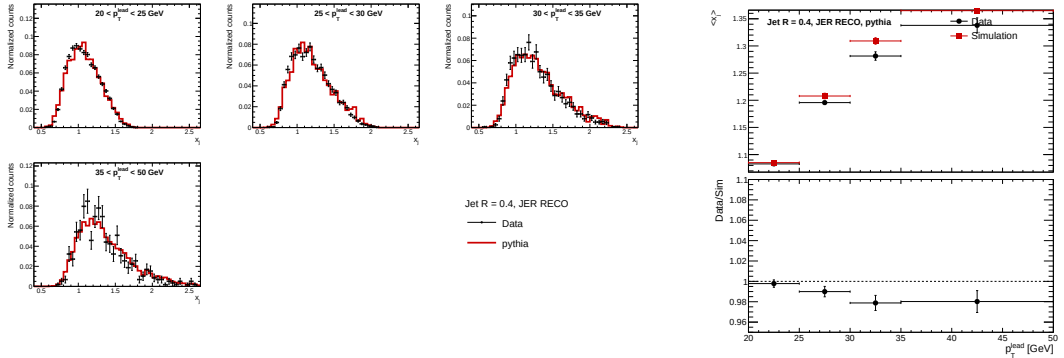
- ▶ Data lies 2–3% above MC in every bin.
- ▶ The MC x_j distribution peaks slightly lower than data.

Configuration 2: individual smearing, recoil ≥ 14 GeV



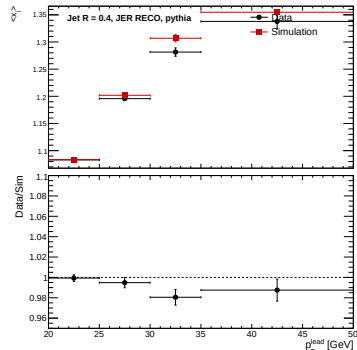
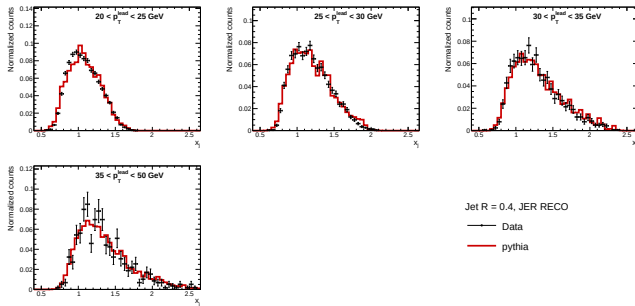
- ▶ Nearly identical to configuration 1.
- ▶ Two jets of at least 7 GeV almost always give a recoil above 14 GeV.

Configuration 3: add unsmeared, smear the recoil, recoil ≥ 14 GeV



- ▶ The distributions agree over the whole x_j range in the first three bins.
- ▶ The ratio is about 1.00 at 20–25 GeV and 0.98–0.99 above.

Configuration 4: recoil smearing when jets 2 and 3 are one truth jet



- ▶ Jets 2 and 3 from one truth jet: added unsmeared, the sum smeared at that truth jet's p_T .
- ▶ All other events: jets 2 and 3 smeared individually.
- ▶ The ratio is 1.00 at 20–30 GeV and 0.98–0.99 above.

Summary

- ▶ How the recoil is smeared, not the recoil cut, sets the data/MC agreement.
- ▶ Smearing jets 2 and 3 individually leaves a 2–3% data excess.
- ▶ Adding jets 2 and 3 unsmeared and smearing the sum once removes it.
- ▶ Adopted: configuration 4, with truth jets ≥ 7 GeV.
- ▶ Its data/MC ratio is 0.999, 0.995, 0.981 and 0.988 in the four bins.
- ▶ Configuration 3 agrees within 0.8%, and the hybrid shows the width choice plays no role.
- ▶ The 5 GeV truth-jet threshold moves the ratio by up to 0.009, a candidate systematic.